

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA11

COURSE OUTCOME COVERED

***CO1:** Apply object-oriented concepts and software development life cycle models to design structured and modular software systems. (BL2, BL3)*

Assessment Tool Weightage: 60%

Dr Ramamoorthy

QUESTIONS: 12

Total Marks:60

Annexure A

CONCEPT MAPPING

Code of Conduct:

I B. Harshitha (192420006) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

B.Harshitha

Annexure A

CONCEPT MAPPING

1. Construct a Concept Map

Given:

Requirement Analysis → _____ → Coding → Testing → _____ → Maintenance

Answer:

The missing concepts are:

- Design
- Deployment

Explanation:

- Design converts requirements into system architecture.
- Deployment releases the software to end users.

Analysis:

- Ensures systematic software development.
- Reduces project risks.
- Improves software quality and maintainability.

(5 Marks)

2. Construct a Concept Map

Given:

Object-Oriented Principles

→ Abstraction

→ _____

→ Inheritance

→ _____

Answer:

The missing concepts are:

- Encapsulation
- Polymorphism

Explanation:

- Encapsulation protects data from unauthorized access.
- Polymorphism allows one interface to support multiple behaviors.

Analysis:

- Improves security and flexibility.
- Enhances code reusability.
- Supports efficient software design.

(5 Marks)

3. Construct a Concept Map

Given:

Class → Blueprint

Object → Instance

Method → _____

Attribute → _____

Answer:

The missing concepts are:

- Behavior
- State

Explanation:

- Methods define the behavior of objects.
- Attributes represent the state of objects.

Analysis:

- Helps model real-world entities.
- Defines object characteristics and actions.
- Improves system organization.

(5 Marks)

4. A Concept Map Shows

Given:

Software Quality

→ Reliability

→ _____

→ Maintainability

Answer:

The missing concept is:

- Usability

Explanation:

- Usability measures how easily users interact with software.
- Reliable software performs consistently.
- Maintainable software is easy to update.

Analysis:

- Improves user satisfaction.
- Enhances product acceptance.
- Contributes to software success.

(5 Marks)

5. UML Concept Map

Given:

UML

→ Structural Diagrams → _____

→ Behavioral Diagrams → Activity Diagram

→ Interaction Diagrams → _____

Answer:

The missing concepts are:

- Object Diagram
- Communication Diagram

Explanation:

- Object diagrams represent object instances and relationships.
- Communication diagrams show message exchange among objects.

Analysis:

- Improves system visualization.
- Supports object-oriented analysis.
- Facilitates stakeholder communication.

(5 Marks)

6. Given:

Requirements → Design → Coding + → Testing + → Deployment +

Explanation:

- Failure occurs during the Coding phase.
- Program logic contains implementation errors.

Analysis:

- Errors affect subsequent phases.
- Testing cannot proceed effectively.
- Software quality decreases.

Solution:

- Review source code.
- Apply coding standards.
- Conduct code inspections before testing.

(5 Marks)

7. Construct a Concept Map

Given:

Problem Identification → _____ → Design → Implementation → _____

Answer:

The missing concepts are:

- Requirement Analysis
- Testing

Explanation:

- Requirement Analysis identifies user needs and system requirements.
- Testing verifies whether the implemented system meets the specified requirements.

Analysis:

- Improves software quality.
- Reduces development errors.
- Ensures successful project completion.

(5 Marks)

8. Construct a Concept Map

Given:

Object-Oriented Features

→ Class

→ _____

→ Message Passing

→ _____

Answer:

The missing concepts are:

- Object
- Dynamic Binding

Explanation:

- Objects are instances of classes.
- Dynamic binding allows method calls to be resolved at runtime.

Analysis:

- Supports flexibility in software design.
- Improves code maintainability.
- Enables runtime polymorphism.

(5 Marks)

9. Construct a Concept Map

Given:

Software Development Models

→ Waterfall

→ _____

→ Spiral

→ _____

Answer:

The missing concepts are:

- Incremental Model
- Agile Model

Explanation:

- Incremental development delivers software in stages.
- Agile supports iterative development with continuous customer feedback.

Analysis:

- Enhances adaptability.
- Reduces project risks.
- Improves customer satisfaction.

(5 Marks)

10. Construct a Concept Map

Given:

Class Diagram

→ Class

→ _____

→ Association

→ _____

Answer:

The missing concepts are:

- Attribute
- Method

Explanation:

- Attributes define the properties of a class.
- Methods define the operations performed by a class.

Analysis:

- Represents object structure effectively.
- Improves software documentation.
- Supports object-oriented design.

(5 Marks)

11. Construct a Concept Map

Given:

Software Engineering Principles

→ Modularity

→ _____

→ Reusability

→ _____

Answer:

The missing concepts are:

- Maintainability
- Scalability

Explanation:

- Maintainability simplifies software updates.
- Scalability enables systems to handle increased workload efficiently.

Analysis:

- Improves software performance.
- Reduces maintenance effort.

- Supports long-term software evolution.

(5 Marks)

12. Given:

Requirement Analysis ✓ → System Design ✓ → Coding ✓ → Integration ✗ →
Deployment ✗

Explanation:

- Failure occurs during the Integration phase.
- Individual modules fail to communicate correctly.

Analysis:

- Causes functional inconsistencies.
- Delays software deployment.
- Increases debugging effort.

Solution:

- Perform integration testing.
- Verify module interfaces.
- Resolve compatibility issues before deployment.

(5 Marks)

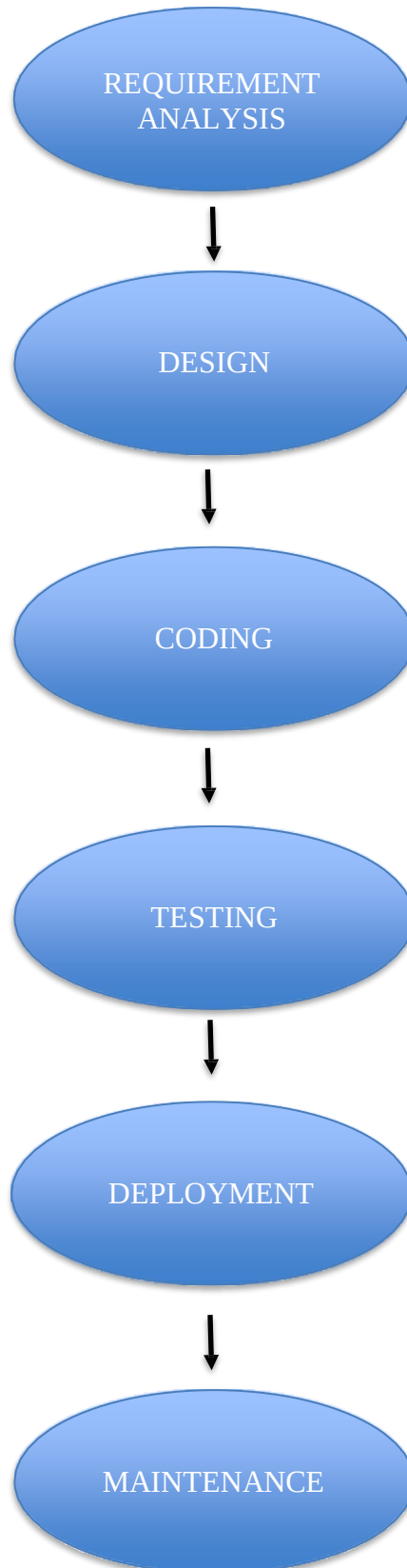
CONCEPT MAPPING

RUBRICS FOR CALCULATION

CONCEPT MAPPING

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

1.CONCEPT MAP



Answer:

- Missing concepts: Design and Deployment

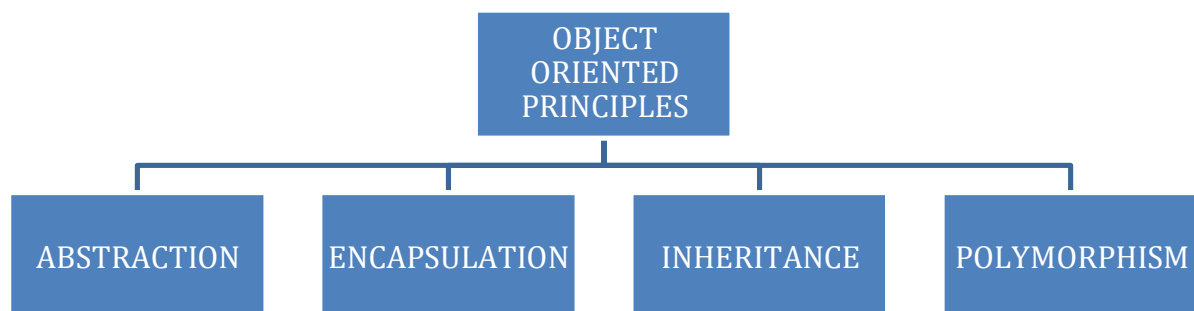
Explanation:

- Design converts requirements into the system architecture.
- Deployment releases the software to end users.

Analysis:

- Ensures systematic software development.
- Reduces project risks.
- Improves software quality and maintainability.

2. CONCEPT MAP



Answer:

- Missing concepts: Encapsulation and Polymorphism

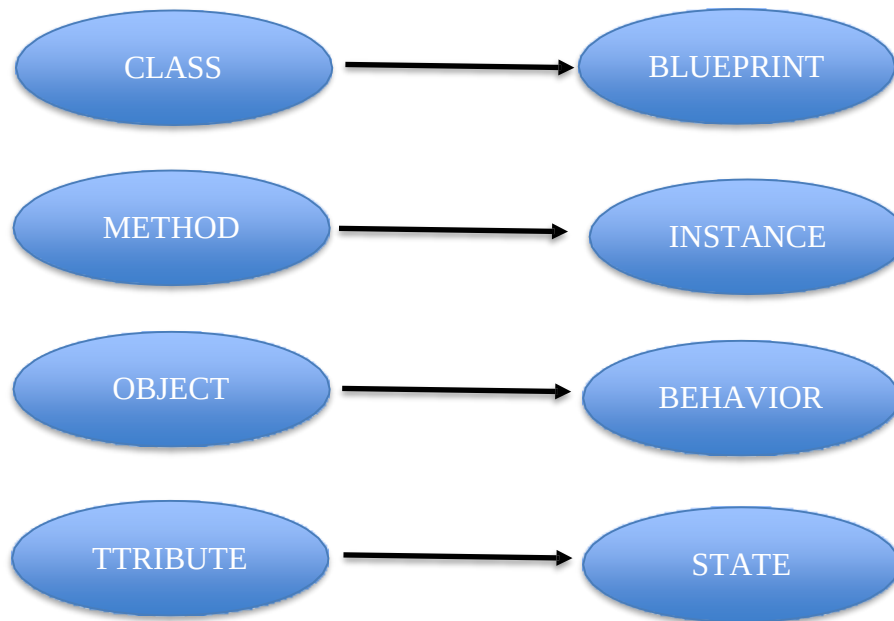
Explanation:

- Encapsulation protects data from unauthorized access.
- Polymorphism allows one interface to support multiple behaviors.

Analysis:

- Improves security and flexibility.
- Enhances code reusability.
- Supports efficient software design.

3. CONCEPT MAP



Answer:

- Missing concepts: Behavior and State

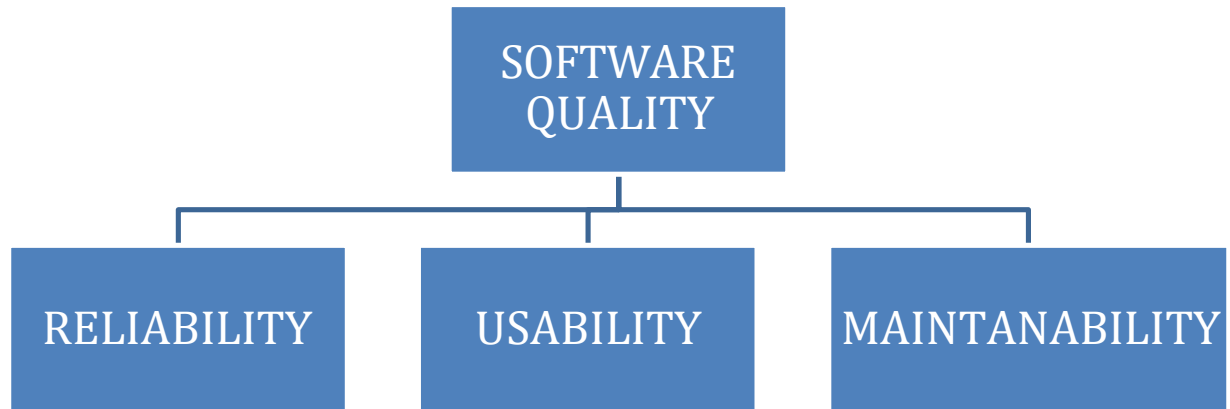
Explanation:

- Methods define the behavior of objects.
- Attributes represent the state of objects.

Analysis:

- Helps model real-world entities.
- Defines object characteristics and actions.
- Improves system organization.

4. CONCEPT MAP



Answer:

- Missing concept: Usability

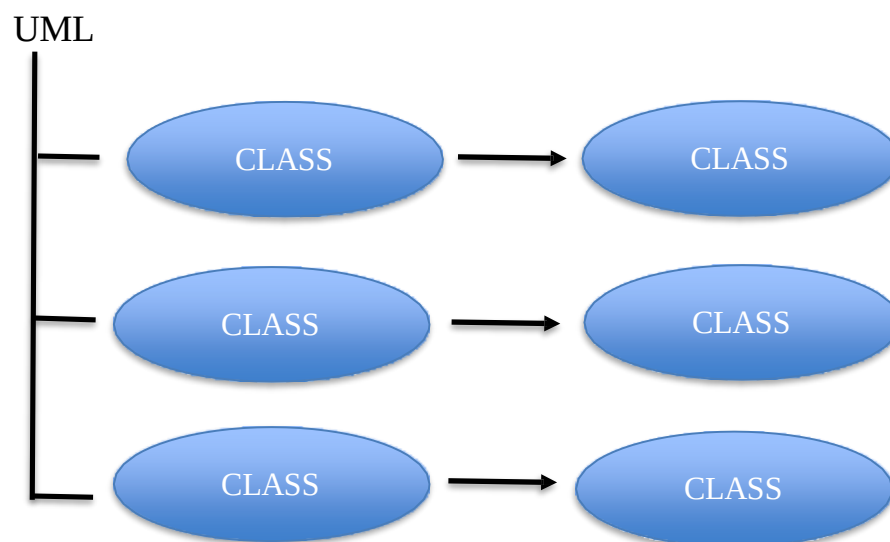
Explanation:

- Usability measures how easily users interact with software.
- Reliable software performs consistently.
- Maintainable software is easy to update.

Analysis:

- Improves user satisfaction.
- Enhances product acceptance.
- Contributes to software success.

5. CONCEPT MAP



Answer:

- Missing concepts: Object Diagram and Communication Diagram

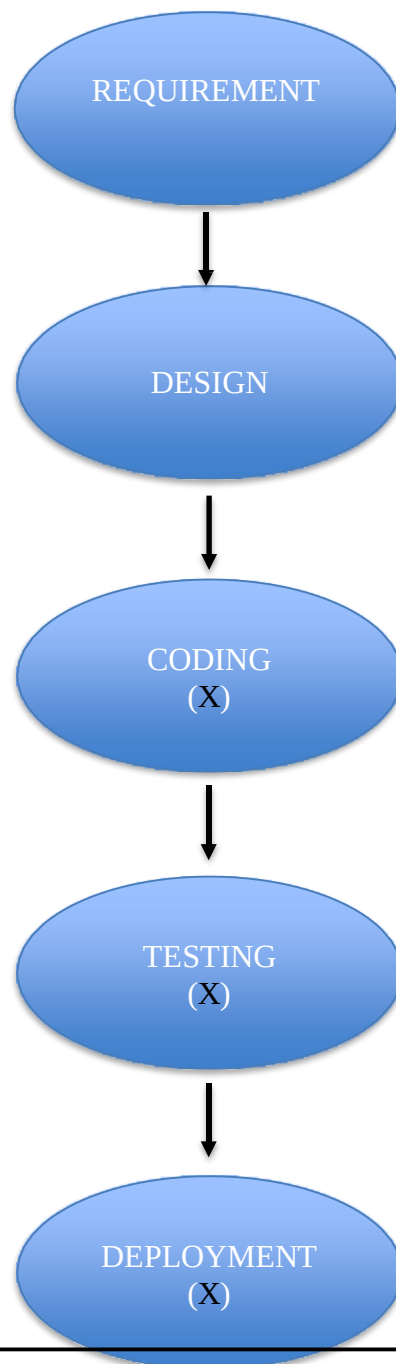
Explanation:

- Object diagrams represent object instances and relationships.
- Communication diagrams show message exchange among objects.

Analysis:

- Improves system visualization.
- Supports object-oriented analysis.
- Facilitates stakeholder communication.

6. CONCEPT MAP



Explanation

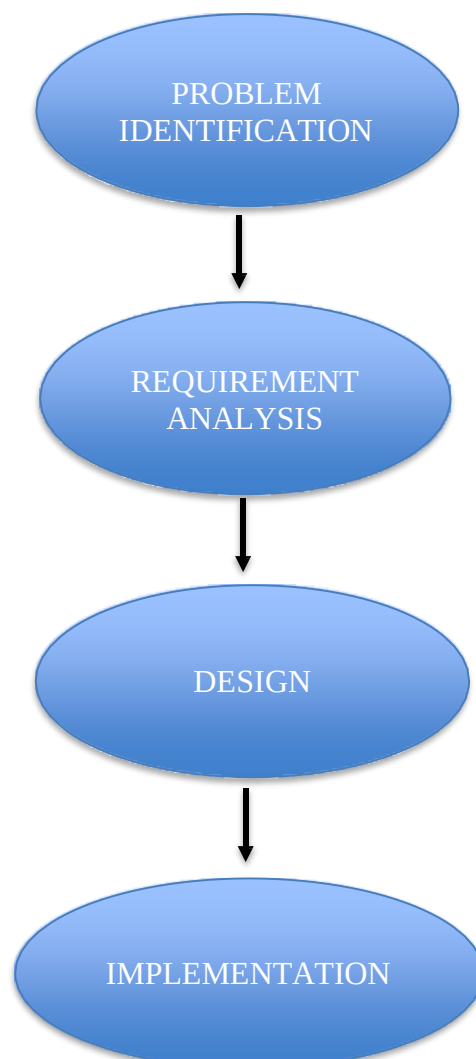
- Failure occurs during the Coding phase.
- Program logic contains implementation errors.

Analysis

- Errors affect subsequent phases.
- Testing cannot proceed effectively.
- Software quality decreases.

Solution

- Review source code.
- Apply coding standards.
- Conduct code inspections before testing.

7. CONCEPT MAP

Answer

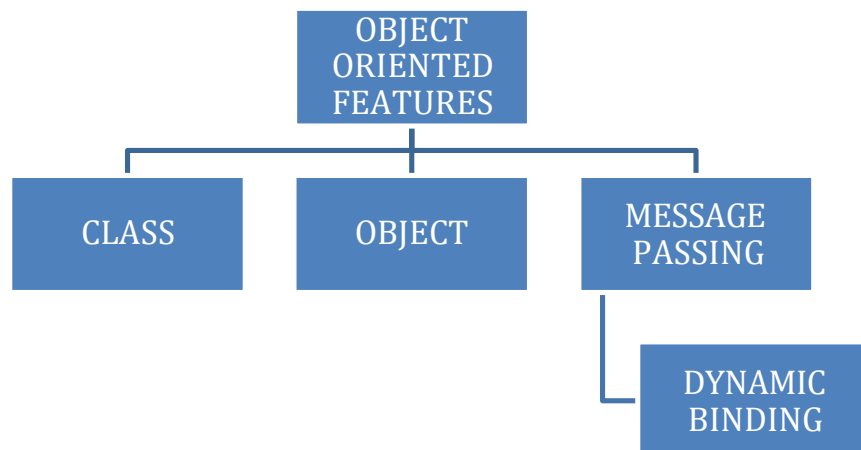
- Requirement Analysis
- Testing

Explanation

- Requirement Analysis identifies user needs and system requirements.
- Testing verifies whether the implemented system meets the specified requirements.

Analysis

- Improves software quality.
- Reduces development errors.
- Ensures successful project completion.

8. CONCEPT MAP**Answer**

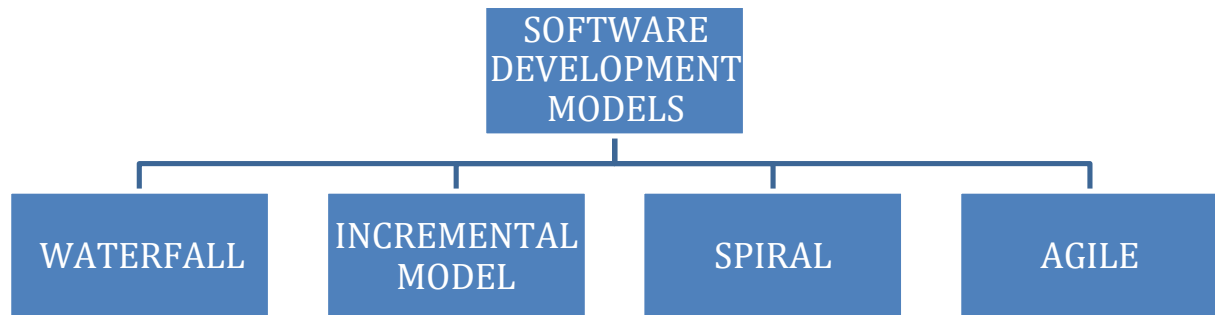
- Object
- Dynamic Binding

Explanation

- Objects are instances of classes.
- Dynamic binding allows method calls to be resolved at runtime.

Analysis

- Supports flexibility in software design.
- Improves code maintainability.
- Enables runtime polymorphism.

9. CONCEPT MAP**Answer**

- Incremental Model
- Agile Model

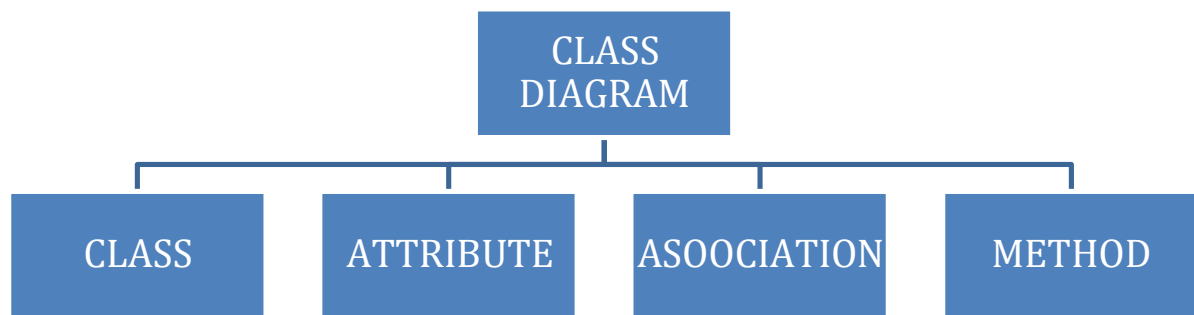
Explanation

- Incremental development delivers software in stages.
- Agile supports iterative development with continuous customer feedback.

Analysis

- Enhances adaptability.
- Reduces project risks.
- Improves customer satisfaction.

10. CONCEPT MAP



Answer

- Attribute
- Method

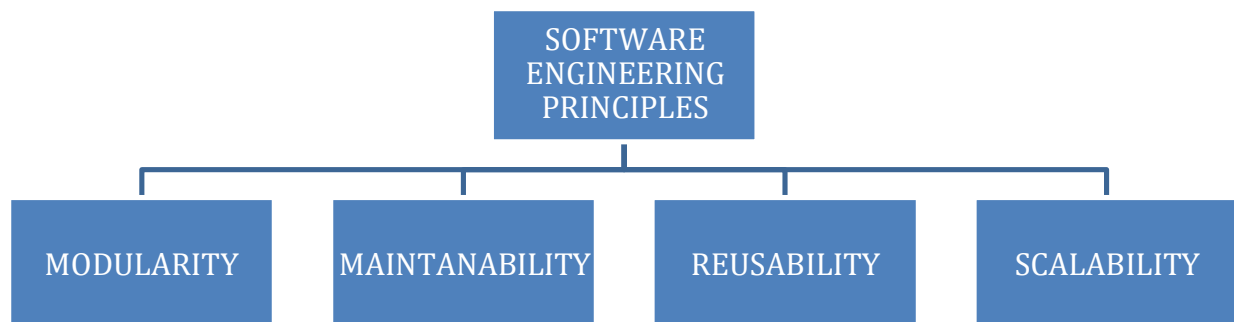
Explanation

- Attributes define the properties of a class.
- Methods define the operations performed by a class.

Analysis

- Represents object structure effectively.
- Improves software documentation.
- Supports object-oriented design.

11. CONCEPT MAP



Answer

- Maintainability
- Scalability

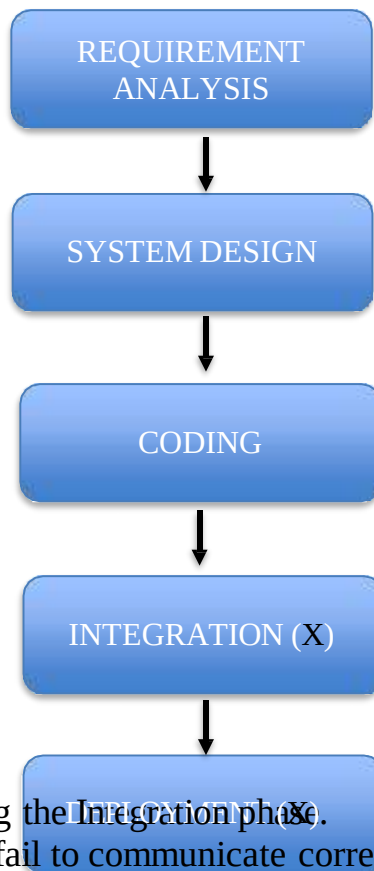
Explanation

- Maintainability simplifies software updates.
- Scalability enables systems to handle increased workload efficiently.

Analysis

- Improves software performance.
- Reduces maintenance effort.
- Supports long-term software evolution.

12. CONCEPT MAP



Explanation

- Failure occurs during the Integration phase.
- Individual modules fail to communicate correctly.

Analysis

- Causes functional inconsistencies.
- Delays software deployment.
- Increases debugging effort.

Solution

- Perform integration testing.
- Verify module interfaces.
- Resolve compatibility issues before deployment.